
How We Contribute

Permaculture Economics for Regenerative Cosmo-Local Networks

Roberto Valenti & Jillian Hovey

Please note: the authors are from Italy and Canada, and this article uses a mix of European/British, Canadian, and American English.

Foreword by Jillian Hovey

Back in 2000, I was teaching on a Permaculture Design Course at the Bullock Brothers' on Orcas Island. I was in the teaching building, which was situated on the keyline of the sloped property, speaking at the front of the room to the entire class, and I remember something like this spontaneously coming out of me:

"We need all of us to practice permaculture in our own ways. Every one of you is on a path to find how YOU will practice permaculture. Ground yourself in the fundamentals of permaculture, and then be on a journey - listening and feeling your way to your unique expression of permaculture. Permaculture is practiced through the "instrument" of you. The world needs you. It needs ALL OF US to practice permaculture in a wide diversity of ways, as that is the only way we will ever begin to be able to do the scope of regeneration work that is needed. This needs to be a way of life for us all, and you following your path will help move us towards that goal."

I had never said that before - it just came out. It felt edgy, as I had never heard permaculture described that way before; but it felt true. And over the years since, it has felt increasingly true as the world has become readier to see and practice permaculture that way.

"Diversity" is a core permaculture principle. Why then have we not been open to a diversity of ways in which permaculture is understood and practiced? Since I formally got involved with permaculture in 1996, I have seen a lot of subtle and not-so-subtle ways in which people were shut down for wanting to pursue permaculture in ways that were different from the dominant land-care and teaching practices. We need to evolve beyond that if we want to co-create permaculture movements for regeneration in our homes, local areas, bioregions, countries, continents, and the planet. When we create space for other ways of thinking, seeing, processing, valuing, practicing can come in. We need ALL OF US to practice permaculture in a DIVERSITY OF WAYS.

Change has been slow, but has been happening. Over the past few years, we have seen the advent of "Social Permaculture," which has been lifting up the "human ecologies" in permaculture. This helps us to continue to grow a wider frame beyond the very strong influence of the rational scientific view of the environment being a thing "out there," with the human observer being invisible. We need to understand the needs and yields of humans, and take them into account if we want our

designs to be more real and effective.

Ultimately, if we want to co-create regenerative solutions in our ways of life, I think that we need to inspire a vast, global webwork of people motivated to practice permaculture in a diversity of ways. Indeed, that was always the intention of Bill Mollison (permaculture's co-founder). Limiting our understanding and practice of permaculture will not get us to the global regenerative response that is needed. We need all of us!!

Wholistic Permaculture Practice in the Global Community

In my wholistic permaculture practice (*please note that I spell wholism with a "w" on purpose, as I feel it represents more to the "whole" that I am invoking*), I have traveled and worked with a variety of people and projects around the world. I support a lot with forming and existing eco communities, which tend to be more aware of, and committed to, the social aspects of our work together. But even though those communities may be ahead of the curve, all projects I have worked with could stand to be more skilled in the realm of "human dynamics." As a result, for many years, my permaculture practice has included literacy in human ecologies: self awareness and personal growth; healthy relationship skills, such as effective communication; decision-making processes; etc. I have seen the positive impact that developing more capacity in those areas can bring to what we hope to achieve in the world.

Though I myself have been ahead of the curve (which was not an easy path), I have been glad to see that steady increase in acceptance of integrating social awareness and healthy practices into permaculture over the past decades. For the past couple of years, I have seen a new pattern of integration with permaculture: this time, it is with the global "Web3 Technology." There is a growing community of awake tech people who are motivated to bring their tech skills to help regenerate the planet; and, many of them are also working on their self-awareness, personal growth, communication skills, physical health, etc.

I have been working more and more with these types of people and groups since my time at the Synergy Hub in Rotterdam in 2016. We were living in a decommissioned old age home (which was actually a great infrastructure for our community to inhabit!). There were people of all ages, from a variety of organizations, such as United Earth, the School of Social Alchemy, NooMap, and Yunity (the radical food-sharing

group). There was also a tech wiz mining cryptocurrencies in the basement, who went on to become a core coder for Holochain (*a post-blockchain agent-centric framework*). I was inspired by the holonic work of NooMap, and began to see the potential of tech supporting our permaculture work. Indeed, “Appropriate Use of Technology” is one of the core original principles of permaculture (*before the 12 principles later developed by David Holmgren*).

Trip to Liminal Village

In the fall of 2021, I traveled to Italy for a design session for a global connection platform called “Co:Do”. It was held at Liminal Village, which is an eco-tech hub in the center of Italy. It had been created by the vision of its founders, Roberto Valenti and Laura van Wijngaarden. I spent time with them throughout the design process - looking at their land, and discussing their dreams. I stayed on and worked further with Roberto and Laura after most of the other folks left. In that process, I shared some fundamental permaculture tools with Roberto, which he really took onboard.

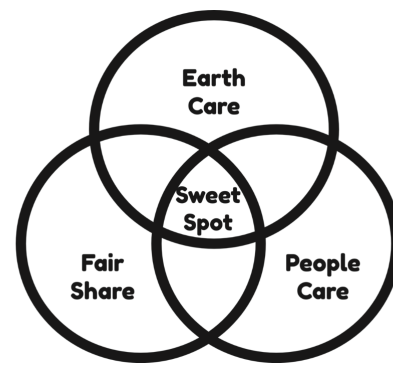
Roberto and I continued to work together after I returned home to Canada in early November. Before Christmas, Roberto invited me to join an online Winter Solstice gathering with some other members of his eco-tech collective in Europe. Connecting with people at that event led to Roberto and I developing a slide show presentation for an annual eco-tech conference that is held between Christmas and New Year’s. Roberto took the lead in the development of the presentation, and I was pleasantly surprised to see how he focused on the permaculture tools and showcased how they could be used in his regenerative economic vision for the world (*and how to code for it!*). People were inspired by our 2-hour presentation and workshop, and I was happy to inoculate a group of tech people with permaculture. I trusted it would weave its way into their thinking, and somehow contribute to our collective evolution - especially with tech people who are actively working towards solutions!

Over the past year, Roberto and I (*and his wife Laura, and our colleague Robert Schram*) have given a few other presentations. It has been rewarding to see the application of permaculture grow and evolve. Now, for the first time, Roberto and I have moved into the realm of writing an article together about these topics. I think that Roberto has a compelling vision, which I felt was a good fit for this “How We Contribute” issue. I think and feel that he has something to bring forward that will help us evolve.

From this point onwards, there is a blend of Roberto’s and my writing, and the remainder of this article will be in a shared voice. Enjoy!

Introduction: Permaculture As a System of Ethics

As human beings, we naturally seek ways to make a positive impact in the world around us. This is life-force energy in action. Permaculture is a wholistic design philosophy that seeks



The intersection of the three Permaculture’s ethics is the “sweet spot.”

to create human systems that align with that energy. The aim of permaculture design is to create sustainable human settlements and regenerate damaged ecosystems, while meeting healthy human needs.

Permaculture is more than just a set of principles or techniques: it is also a set of ethics that guide our actions and decisions. These ethics include care for the earth, care for people, and care of the future (*which can also be known as “fair share”*), and they are foundational to the permaculture approach to design and sustainability.

The three permaculture ethics are traditionally represented as the intersecting circles of a Venn diagram. Ideally, we are aiming towards the “sweet spot” of the intersection of the three overlapping ethics. Ultimately, the three circles could move towards increasing overlap, so that we have a deep integration and practice of all three ethics at the same time. Never-the-less, these permaculture ethics give us three different valid lenses on the whole. If we would apply these ethics to our economic system, each ethic would suggest the following aspects:

Earth Care: protecting and restoring the natural world. Our economic system would need to respect ecological boundaries while pursuing social well-being in order to address issues such as inequality, environmental degradation, and the climate crisis.

People Care: supporting the well-being of all people. Our economic system would need to be able to aid in the satisfaction of all psycho-physical needs and wellbeing of everyone.

Fair Share or Future Care: sharing resources and wealth in an equitable and sustainable manner. Our economic system would need to shift away from self-interest and towards pro-social relationships in order to create a viable economy.

In the following sections, which are organized according to the three ethics, we will explore how permaculture ethics, principles, and tools can be incorporated into our lives and our economic systems to contribute to a regenerative, healthy and equitable future for our human communities and the planet.

Permaculture is a design system for creating sustainable human environments. - Bill Mollison

EARTH CARE: The “First” Ethic of Permaculture

A Historical Framework

Earth Care is the ethic with the most developed permaculture practices. The main focus of permaculture has always been on land-based practices, and most people associate permaculture with gardening or farming (“permanent agriculture”). During the time that permaculture has been popularized around the world, people trying to make their living at permaculture have predominantly engaged with it as landscape design consultants and/or as permaculture teachers.

This pattern is evident when people are inspired by a Permaculture Design Course (PDC), and want to learn more and further their education. The main “advanced” education option has usually been a Permaculture Teacher Training; and occasionally, there are advanced design courses, which have almost always been landscape-focused. As such, there has been a very strong framework in permaculture around land-based practices, and that the people who want to develop or progress, become teachers or landscape designers.

From Sustainable to Regenerative

Most of us who come to permaculture are motivated to live in more sustainable ways. These days, “sustainable” has been supplanted by the term “regenerative,” because simply “sustaining” the status quo is insufficient to steer us away from the collapse of our human and natural systems. We need to regenerate systems at all scales. Fundamentally, permaculture is about co-creating a webwork of regenerative relationships that benefit all life and their environments. By learning to weave this web, we have the capacity to do extraordinary things, such as regenerate deserts into thriving ecosystems.

Permanent Culture

While land-based permaculture is hugely important, even if we did just focus on “Earth Care,” we are seeing that we need to include socio-economic ecologies if we want to be effective in catalyzing the regenerative response capacities of permaculture in the world. A wholistic “permanent culture” approach is likely more of what is needed. We can take the same approaches that are used to study and understand nature, and apply them to human ecosystems.

Only focusing on “Earth Care” has limited the scope of how people understand and practice permaculture. Permaculture was always intended to be a wholistic approach to living in harmony with the intelligence of nature, as is indicated by the scope of the “other two” ethics of People Care and Fair Share. Most of this content about this is in the last chapter (14) of *Permaculture: A Designer’s Manual*, which is entitled “Strategies of an Alternative Global Nation”. In Chapter 14, the focus is alternative societal structures, such as legal and economics.

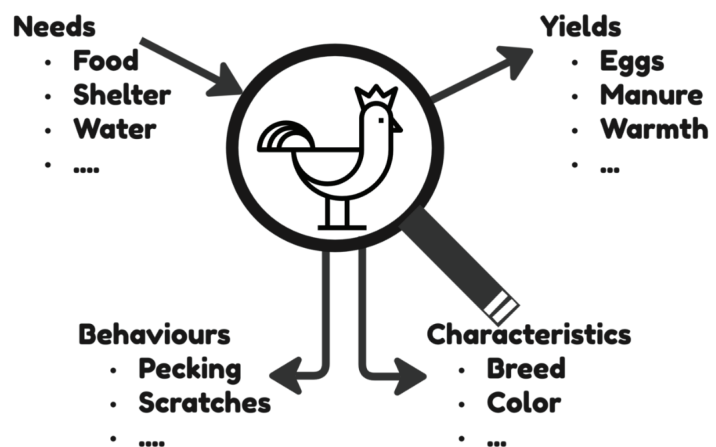
As we move beyond this Earth Care section of this article, and get into the “People Care” and

“Fair Share” sections, we will explore those topics more. But for now, let’s take a look at a classic permaculture tool that we will also use in social systems.

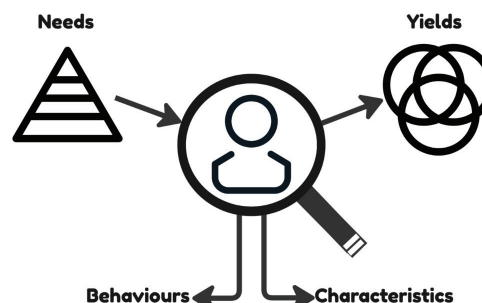
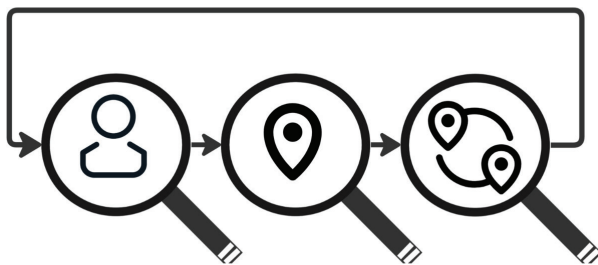
Permaculture as System Thinking Tools

As stated earlier, permaculture is about creating an interconnected webwork of regenerative “needs and yields.” While “needs” can be fairly readily understood as what we need to live, for our purposes, “yields” means what we give out to the world - what the “output” is. Permaculturists are meant to study and facilitate that web of needs & yields, so it is useful to consider *how* to do that. One way is through the permaculture practice of analyzing an “element” in our permaculture systems. This is a powerful foundational permaculture tool, which is commonly known as the “Needs & Yields” analysis (or “Inputs & Outputs”). It helps us to see more into the Needs, Yields, Behaviours & Characteristics of an “Element” in a system. This process is akin to using a magnifying glass (or “lens”) to examine any part of a system.

To illustrate this tool, we can use the classic example of the “Permaculture Chicken,” where we analyze its Needs, Yields, Behaviours and Characteristics. This exercise supports us in seeing the living dynamic of an element in more detail, primarily through what it needs and what it yields. This aim is to facilitate healthy connections between the chicken and other elements, where we connect the chicken’s yields (like heat, meat, eggs, CO2, alarm, companionship, pest control, etc.) to the needs of other elements in the system. And, we also want to meet the chicken’s needs (*shelter, water, food, companionship, disease control, etc.*) from the yields of other elements in its environment - either elements that are there already, or ones that can potentially be manifested there. The Needs & Yields analysis tool fosters system thinking in our design work, which is a big part of what we are after in permaculture design.



Using the permaculture “lens” to study Needs & Yields (or “Inputs & Outputs”), as well as the Behaviours & Characteristics of an element in a system



A simplified version of chaining the Needs & Yields lens applied to larger entities, such as people, locations, and the network.

The Needs & Yields lens applied to humans

Applying the Permaculture “Lens” to Any Element in a System

This Needs & Yields practice regards any entity in a system as an element which we can analyze using this tool. In traditional land-based permaculture design, beyond chickens, we look at other landscape elements such as: fence, roof, pond, cow, etc. Beyond single elements in a system, the permaculture analytical lens can also be applied to any entity, be it “simple” or complex, like “house,” which is composed of many elements and sub-systems. This skill of understanding the needs and yields of an element helps us to see more possible connections between entities in any given system, such as people, places, projects, and even the network connecting them.

As permaculture designers, it is an essential skill to become more literate in the inputs (*needs*), outputs (*yields*), and intrinsic behaviours & characteristics of system’s elements. It puts us on the path to becoming adept at facilitating connections and creating that webwork of relationships, which is the bedrock of the “magic” of permaculture design. As we will see in the next section, which is focused on the “2nd” permaculture ethic of People Care, we can apply this Needs & Yields tool to our human landscapes.

PEOPLE CARE: Permaculture Tools Applied to Human Ecosystems

“I’d come into town from the bush - after 28 years of field work in natural systems - and become an academic. So I turned my attention to humans, much as I had to possums in the forests.” - Bill Mollison, Permaculture Design for Living.

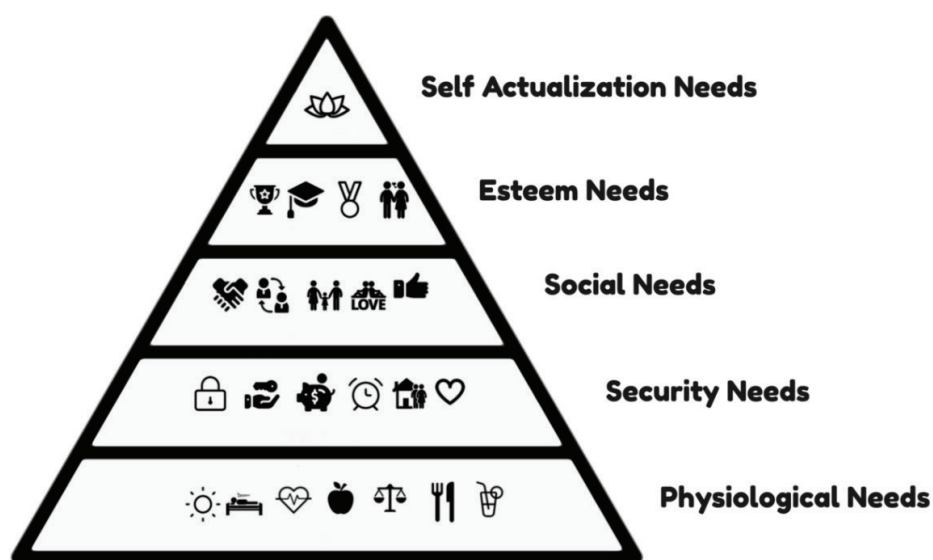
Applying the Permaculture Lens to Humans

As permaculture expands its scope to more overtly social realms, we can find some powerful benefit in using the classic land-focused Element Analysis tool, and applying it to our human ecologies. In fact, as we can apply the Needs & Yields analysis to *any* element of an ecosystem, we might want to turn the magnifying glass on ourselves. Awareness of what each “element” in a system needs is critically important to helping to co-create systems to meet those needs in regenerative ways.

Human Needs

Human needs are strong drivers in our landscapes, and affect almost everything we hope to do as permaculturists. In our modern culture, we have been trying to meet “not necessarily healthy” human needs in “not necessarily healthy” ways, which has caused deep dysfunction in our world. A very leveragable strategy for becoming more regenerative would be to *meet healthy human needs in healthy ways*.

A classic, widely-known model to understand human needs is Maslow’s Hierarchy of Needs, which is represented as a pyramid.



Maslow’s Hierarchy of needs - in pyramid form, with “base” needs at the bottom, and building towards our highest potential.

mid. The basic human needs form the bottom of the pyramid and are called the “Physiological Needs.” The next layer up is “Security Needs.” That transitions to the middle layer, which is “Social Needs.” That is followed by the “Esteem Needs.” And the apex of the pyramid is “Self-Actualization,” which is described as achieving one’s full potential.

Maslow’s model offers a structure that helps us see our human needs more clearly - especially some of the “higher” needs, which can be assumed or implied in our cultures, but not necessarily talked about. This structure can be also used to inform our design process: for example, we might want to first focus on designing for basic needs on site. Meeting those basic needs then creates more opportunity for needs higher up the pyramid to be met. This dynamic creates the conditions for us to move in succession up Maslow’s hierarchy of needs as a system evolves and matures.

Human Yields

One of the classic “Mollisonisms” of permaculture, is that “Everything Gardens.” That is a way of saying that everything that we do in the process of being alive creates the conditions for growth for other entities in a system. What we humans “do” in the world - or, how we “be” - has effects on the world around us, others, as well as ourselves. “Everything gardens” is also a way to see the activities that result in what humans “yield.” You could say that “how we garden” is how we contribute in the world. Human have a wide range of “yields”, from intentional to un-intentional, natural to un-natural, good, bad, beautiful, ugly, etc. The possibilities for human “yields” are almost endless, and those possibilities can increase when we create favorable conditions for yields to thrive.

Relationships

Our human yields are how we contribute to the world - by being, as well as doing. We can give to one another in relationships, with the yields from one fulfilling the needs of another. Permaculture is about facilitating that dynamic, by connecting needs & yields in the widest range of relationships. The possibilities for connections of the yields of humans to the needs of other humans, as well as to other elements of a system, are essentially infinite. *(Remember that the possibilities of design are only limited by the creativity and imagination of the designer, which is another Mollisonism).*

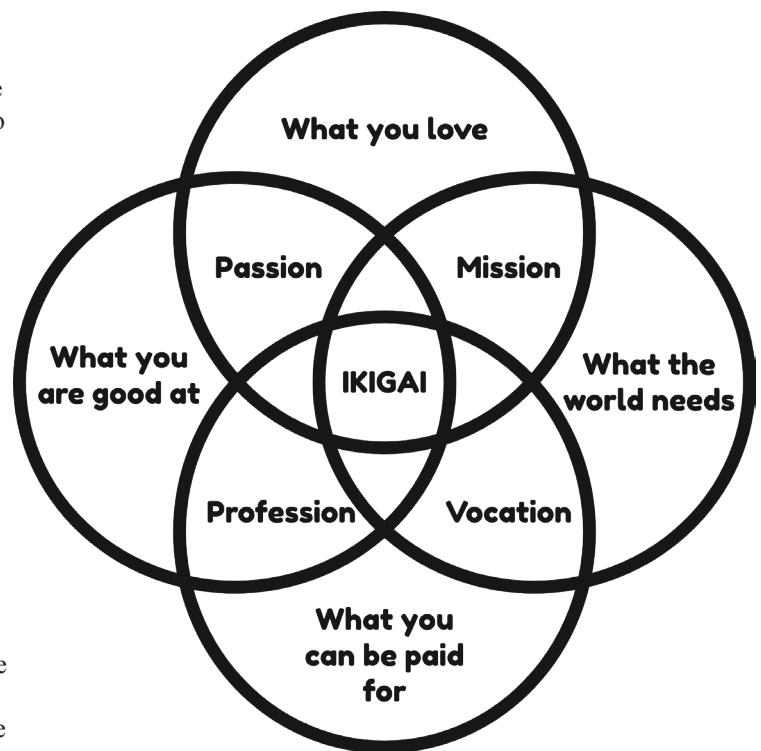
But, do we even know what our needs are? Some of them are obvious, while others are socially opaque. Indeed, many of our deepest needs are not socially acceptable to talk about, and some are even “taboo.” Our literacy in human needs has often been inhibited by society, but that has been changing over time. We need to continue to become more knowledgeable about human needs, and how they can be connected with human yields, if we want to stop degrading the planet. That is because our voracious human appetite is destroying the planet, and our future ability to sustain life for all. What would it be like to get all of our needs met in regenerative ways?

Ikigai

The “Ikigai” model can help us see deeper into human needs and yields, and inform us in our understanding of how we contribute to the world. Ikigai is a Japanese term that translates to “reason for being” or “reason to live.” It’s the idea that everyone has a purpose or calling in life that brings them joy and fulfillment. When we use our skills and talents to make a positive contribution, we can help to create the conditions for others to make positive and regenerative contributions, which can lead us to a deeper sense of fulfillment (*which goes well beyond just earning a paycheck!*). Ikigai can help us get clearer about what our highest and best contribution could be.

The Ikigai term was taken up by Marc Winn and integrated into a model of “purpose” that has four key overlapping questions:

1. What do you love?
2. What are you good at?
3. What does the world need from you?
4. What can you be paid for?



The Ikigai model, a Japanese concept meaning “the reason for being.”

In this composite model, “Ikigai” is at the central intersection of this Venn diagram, which represents a balance of the 4 aspects. The partial intersections of two of each of these circles define: Vocation, Profession, Passion and Mission. For example, many permaculturists practitioners live at the intersection of “What the World Needs,” “What You Love,” and “What You are Good At,” but do not overlap very strongly into the realm of

“What You Can Be Paid For”!

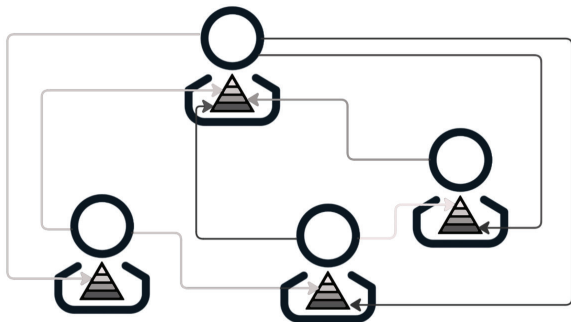
If we analyze this model deeper, we realize that two of the aspects of the Ikigai (“*What You Love*” and “*What You Are Good At*”), can be considered as what you do or “yield.” The other two are ultimately about “needs” - one is the meta construct of “What The World Needs,” and the other is “What You Can Be Paid For”, which is essentially a means to meet your needs. And, further to that last point, “What You Can Get Paid For” could be re-stated as “What You Can Get Access To or Recognized For,” and access and recognition does not just come through money.

Collective Ikigai

Finding one’s personal *Ikigai* can be very challenging, particularly for people who have few resources and little collective assistance. But, while seemingly centered on an individual person, *Ikigai* can also be brought to life in a collective context. Indeed, one aspect of collective *ikigai* relates to the many ways in which a collective can help an individual. While an individual can contribute from what they are good at to the collective; a yield from the collective to the individuals, is to help people in their need to find their unique *Ikigai*. So, the relationship is mutual (“*mutual synthesis of organism and environment*,” which are the words G. Scott Williamson used to define “*health*”).

But there is another aspect of collective *Ikigai*, which is about the way that collectives as a whole find their own “*niche*” in whatever ecosystem they participate in. This mirrors the natural situation where organisms live in groups, such as herds, families, packs, troupes, and also in fields or forests. The individual organism has a role in its group, but also the group as a whole plays a part in the ecosystem in a much more significant way than is possible for an individual.

The long history of natural evolution allows species to grow into ecological niches in a kind of collective *Ikigai*. Human collectives have the potential to do this kind of fitting together through human communication processes. The possibility of collective *Ikigai* can be explored in the contexts of ecovillages, and also of regenerative and commons-oriented business. (For more information, please see our colleague Simon Grant’s article: <https://www.simongrant.org/d/2021-11-30d.html>.)



People Care: Inside a guild, different needs are met by the yields and resources of others.

Guilds: Economies of Needs & Yields

What if we all came into agreement about working together to fulfill each other’s needs? Groups of people could consensually come together to form mutual support “guilds” where they collaboratively work towards meeting each other’s psycho-physical needs. These human guilds could commit to give to each other the safety, the recognition, the love, and the self-actualization that every individual needs.

These guilds would facilitate interconnections between the needs and yields of various entities in them (*e.g. people, places, organizations, etc.*). A network of guilds could aid each other with resources and support to achieve our shared goals more effectively. The myriad connections of those networks would give us resilience far beyond that one guild that we may be involved in.

How can we help these local and global guilds to coordinate to become a self-supporting and replicating ecosystem? Let’s answer this using the third permaculture ethic of “Fair Share” or “Future Care.”

“We ourselves are part of a guild of species that lie within and without our bodies. Most of nature is composed of groups of species working inter-dependently.” - Bill Mollison, Permaculture: A Designers’ Manual

FAIR SHARE: Permaculture Tools Applied to the Ecology of Economics

“The only ethical decision is to take responsibility for our own existence and that of our children.” - Bill Mollison, Permaculture: A Designers’ Manual

Degenerative Human Economies

Fair share means the equitable distribution of resources so that we can meet the needs of all members of our communities. As we face the pressing issues of social inequality and environmental degradation, it has been becoming increasingly clear that our current economic systems are not working for the betterment of all people, nor for the planet. As permaculturists, we desire a world where people care for one another and the earth in equitable ways, yet “Fair Share” is by far the least developed of the three permaculture ethics.

Our current economic system is based on scarcity, competition, and extraction, where most people struggle to stay afloat. As such, it is not conducive to Fair Share. We currently coordinate our collective actions using a monetary system, which is driven by the imperative for profit. The true costs for the environment are not taken into account. As such, the pursuit of money often justifies environmentally destructive actions.

Beyond negative environmental consequences, our monetary economy also undermines our social fabric by promoting individualism. This happens in many ways, including giving the illusion that we do not need each other because we can simply purchase the goods or services we need with money. And though we can indeed make use of money to meet our basic needs, it cannot be used for the “higher” needs for love, belonging, esteem, and self-actualization. In fact, believing we can use money to fulfill these higher level needs precipitates a lot of dysfunction in the world.

As the economic problems are systemic, the solutions will also need to be systemic. As our current systems crumble under their own weight and make space for new ones, their decaying parts can become humus from which we can build an entire new human ecosystem focused on harmoniously meeting the needs of people, the planet - both right now and into the future generations.

As permaculturists, we have the responsibility to deeply change the way we look at the relation between nature and our economic system. Rather than seeing it as being detached from nature, we would need to see our economy, or global coordination system of our needs and yields, as an integral part of nature itself. There is a need and opportunity for a new operating system for the planet - a permaculture operating system, which can support a permanent culture. Let's look to nature for some ideas of how that could work.

Nature's Regenerative Economy

Nature is the archetypal example for regenerative self-sustaining, self-adaptive, and self-replicating systems. If we had an economic system that was based on natural systems, we would probably not be in the situation where we are degrading the planet and experiencing such a great social disparity. As humans, we have the opportunity to apply our intellect to create more efficient and regenerative economic systems which are based on the wisdom of nature. We can use biomimicry (the practice of mimicking the principles of nature) to learn from nature, and transfer natural processes to our social systems.

"Permaculture is a philosophy of working with, rather than against nature; - Bill Mollison, Permaculture Two

Free-Flow Resource Exchange

Let's consider a forest as a reference ecosystem. It does not mint or use tokens to facilitate exchanges; it doesn't make deals, nor does it pay for services, and there isn't a central bank or treasury that distributes the currency and keeps track of how much anyone has. A tree doesn't need compensation from the soil organisms for shedding its leaves, and the soil doesn't put conditions on the amount of resources it makes available back to the plants. As the system co-evolved as a whole, exchanges happen within the entire ecosystem just by "being a part" of that ecosystem; and capacity for meeting needs increases as the whole ecosystem grows through stages of succession.

Accelerating Succession

"Accelerating Succession" is a commonly known catchphrase in permaculture. An interesting take on that is to consider the term "success." Ironically, in our industrial societies, "success" and "successful" usually relate to financial gains. Ironically, that may be the opposite of what is needed, as a focus on accumulation of financial wealth tends to drive the degradation of our environment, and the impoverishment of people and communities. "Success" in permaculture terms is about being effective at co-creating the conditions for yourself and others to meet our needs in healthy and regenerative ways; and as doing that feeds and grows our capacities for all to thrive.

The Gifting Economy

Attempting to integrate nature's solutions with human systems, asks us to see the world as one big inter-related system. Everything we do has an effect on our environment, which means that our actions have consequences and responsibilities. The term that comes closest to how nature does it is a "gifting economy", where resources are not commodities to be bought and sold, but are gifts to be shared with those around us.

In a gift economy, individuals exchange gifts without the expectation of receiving something directly in return for that action. This fosters acts of kindness and generosity, as the absence of a perceived obligation to reciprocate allows individuals to engage in benevolent and selfless acts towards others. Examples of gift economies, where the needs of a community are met by exchanging goods and services without the direct use of money, are: potluck dinners, time banks, barter networks, carpooling networks, and tools libraries.

Another example of the gift economy are community gardens, where members come together to grow food and share it among themselves. These gardens serve as a local source of fresh, healthy food, and foster a sense of pride and investment among members of the community. The gardens promote self-sufficiency, food security, and social connectedness, while also reducing the environmental footprint of food production. Members of the community may contribute with their available time, labour, and resources, but they do not exchange money or other forms of payment. Instead, they gift their labour (*and the fruits of their labour*), to one another. And in doing so, they co-create an ecology that meets some of their physiological and psychosocial needs. If we are part of the same ecosystem, we can be confident that our gifts will somehow increase the capacity and resilience of the ecosystem as a whole, and thus we ultimately all benefit.

Abundance

A tenet of permaculture is to work with the inherent abundance of nature. And indeed, one way that Fair Share is taught, is that we can give from the excess that can come from being intelligent stewards of nature's abundance. That abundance, or "excess" of yield, feeds the many aspects of the system, directly and indirectly. Permaculture is about rich, vibrant, diverse systems that can resiliently meet a variety of needs, without just taking from the environment. If we all contributed to abundant systems and shared yields with each other, there could be more than enough for everyone.

"Security can be found in renunciation of ownership over people, money, and real assets; to gain, keep or protect that which others need for periods of legitimate access. A lending library enables people to help themselves to information; a locked-up book collection is useful only to the person who owns it." - Bill Mollison, from Permaculture: A Designer's Manual

Cosmo-Local Guilds

So, should we just start gifting everything we have to everyone? Our dominant culture has an extractive mindset, is based

in monetary transactions, and is based on the myth of scarcity which inspires competition and hoarding, so that is not an easy sell. One idea is to start by offering resources which we have in abundance, or excess (*e.g.: our time, expertise, unused assets and resources, etc.*), to people in our trust network, or to people and projects that we would like to support.

We can form guilds of people working together and gifting to each other to provide for each other's needs while also caring for their environment. These guilds would not be driven by personal benefit, but by the understanding that guild members are inter-dependent, and that our actions affect each other. These communities can experiment, and share knowledge with other guilds. This will build capacity over time in the guilds and in the network, like humus accumulates on the forest floor which grows the mycelial network.

These guilds can work “cosmo-locally” with other guilds in their local area, in their bioregion, and beyond. “Cosmo-localism” combines elements of both localization and globalization. It prioritizes the use of locally-based “heavy” resources (and thus the reduction of carbon emissions associated with transportation), while also allowing for the sharing on a global scale of “lighter” resources such as small parts, plans, expertise and knowledge. This reduces dependence on markets and non-local resources, while fostering greater collaboration and cooperation between individuals and communities. By working collectively in guilds (and guilds of guilds), we can establish self-sufficient and inter-dependent human ecosystems, much like in natural forests.

“I think the world would function extremely well with millions of little cooperative groups, all in relation to each other.” - Bill Mollison, Permaculture: Design For Living

Permaculture's tools aid us in learning from nature and in intelligently stewarding its inherent abundance. Can we also use those tools to create a permaculture economic system that is inspired by Fair Share, and satisfies the needs of the people while it heals and regenerates the planet? It is possible for us to contribute to a novel economic system that can nurture economies in place-based, digitally-connected regenerative guilds of guilds in our own homes, in our communities, throughout our bioregions, and around the world. Let's look more deeply at how existing and future technology can help.

Pro-Social Protocols for People and Planet

How can local guilds of people, embedded in a fair-share “economic desert,” connect with others and evolve into a thriving, self-sustaining ecosystem, such as a mature 7-layered forest garden? By looking at how nature accomplishes that, we realize that we would need some sort of mycelial network, which is the biggest organism on planet earth. Analyzing the mycelial network as an ele-

ment of our system, we can see that it meets its input needs by feeding on the output of decaying organic matter; and its yields of water and nutrients feed a wide range of elements and systems throughout the forest (*even at great distances*). So how can we create a human network that is able to achieve the same feat?

The Digital Mycelial Network

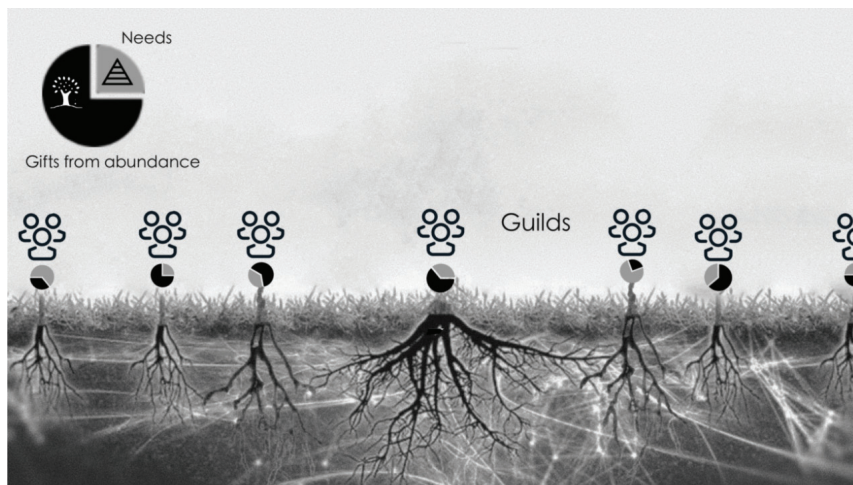
We are in luck, as we already have our mycelial network infrastructure: it's called the Internet! Through the internet, we can already find each other, form intentional guilds, communicate across continents, and share knowledge, solutions, and know how. But, it is only with the latest possibilities from the advent of blockchain technologies (*a.k.a. Web3*), that the Internet can also be used as a peer-to-peer “stigmergic” human coordination system.

Stigmergy is a coordination system present in nature (*see ants*), where individuals leave behind “traces” of their “actions” in the shared environment, which are then picked up and acted upon by others. We would need an agreed-upon stigmergic communications process (or “protocol” in tech speak) that would allow us to optimize the allocation of work and under-utilized resources in our guilds, and connect local needs with local yields. This would make it easier to access and participate in the provisioning of locally-sourced goods and services.

Forward to Nature

Many permaculturists have an almost allergic reaction to “technology” and the internet, let alone lesser-known technologies like blockchain and artificial intelligence (*A.I.*). But, in our modern age, technologies are the means through which more and more of our needs are met. To solve the crisis of our decaying human and natural systems, it may be time we got beyond our prejudices against the “evils” of technology, and come to bless that “unholy marriage” of nature and technology.

Let's not forget: these technologies are also an *element* in our systems, which have yields which we can creatively put to good use to move our civilization *forward to nature* (rather than



Fair Share: needs within the network are fulfilled by abundance of other guilds, using the internet as mycelial network to share the information about the available resources.

“back to nature”).

Now that we have the technologies of internet, blockchain, and A.I., we can upgrade our several-thousand-year-old monetary system into a stigmergic mycelial network, with the following objectives:

- Facilitate the fulfillment of psycho-physiological needs, ensuring wellbeing for all humans
- Facilitate pro-social coordination, favoring co-creation and collaboration over competition
- Regenerate the planetary resource ecology, cultivating widespread abundance
- Remain viable across locations and through time, respecting the local and global boundary conditions of place and planet

Embedding Our Values into Our Technologies

By utilizing blockchain technologies to build our stigmergic protocols, we are able to embed these objectives and values directly into our economic system. This means we can replace the profit motive with a well-defined incentive to serve one another's needs, encouraging producers to orient their behaviour towards fulfilling the community's needs, rather than just their own self-interest. We will also be able to transparently see, and contribute to, any step of the supply chain. This will allow us to be able to consciously choose the paths that minimize the total resource cost (*or the “true price”*).

This gain in visibility of economic activities and supply chain will also allow us to monitor relevant information, like the

carbon footprint of products and services. This would make it easier for both producers and consumers to create circular economic loops and make more sustainable choices. This will help to cultivate “Prosumers”, which are entities which both produce and consume, in connection and balance with others and place.

“The greatest change we need to make is from consumption to production ...” - Bill Mollison, Introduction to Permaculture

There is a long road ahead, but permaculture ethics and frameworks can guide the development of networks of interconnected, value-aligned, and inter-dependent communities. By building and adopting value-aligned, inter-operable protocols on a wide scale, we can provide valid alternatives to our degenerative human systems - alternatives which serve the needs of the people, the planet and of all living beings.

Putting the Ethics into Practice

How We Contribute

We have explored how the permaculture ethics can be put into practice when we are looking beyond the natural systems to human systems. There is much more that we could have discussed in a deeper way, but we feel the most important thing is how we put these ideas in practice. In this last section, we will share how we are endeavouring to bring all of this to life in our daily lives.



Drone view of Liminal Village in Italy, with two community hub houses, olive trees, water collection and purification system, and precision farming.



The “Fiori del Fiobbo” kindergarten and community center, shared Zone 1 of a distributed community of 20 families in Italy.

Connecting the Dots

There are many ecovillages and sustainable initiatives around the globe that have been using permaculture frameworks and contributing to our individual & collective knowledge and capacity for co-creating sustainable human settlements. We are now at a key moment in time, where these initiatives might begin to adopt permaculture-aligned technologies as their mycelial network, and create a new economic “grid” based on the values they share. For this (r)evolution to reach a tipping point, it is essential to create, or find and connect, places that are willing to experiment, educate, and empower individuals and communities in this new mindset. We also need to provide the necessary tools and resources to be able to take action in our respective communities.

A Place “In Between”

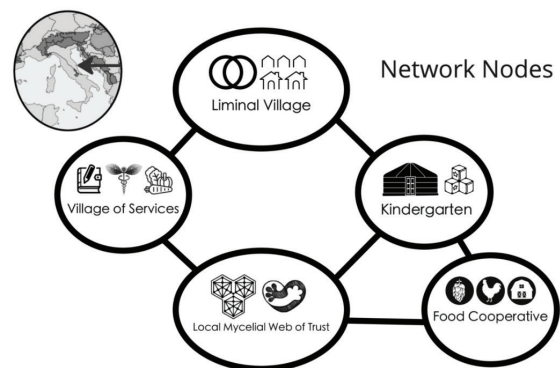
One such place is Liminal Village. It is a co-living hub in rural Italy. Liminal, meaning “the place of transition,” is a cosmo-local hub, where value-aligned people come together to co-create the future they would like to see. Using natural cycles to track progress, participants bring their knowledge and gifts to form temporary communities to craft the tools they sense they need to thrive. There is a strong ethos of intrinsic motivation, so that everyone can move towards their Ikigai. We collaborate in a non-coercive, open space environment, aiming to keep a balance between personal, local and global contributions.

We have been developing and experimenting with open source, cheap and replicable solutions to meet people’s basic

needs (food, housing, water storage and purification), so that people can nurture each other’s higher-order needs, and focus on our truly unique contribution. We aim to use these technologies and innovations to catalyze a local and bioregional network of inter-dependent regenerative hubs and settlements. Motivated people who contribute their knowledge and gifts to the hubs (“What you are good at,” “What you love”), get access to what they need to live a healthy and fulfilled life (“What you can get paid/recognized for”), in satisfying the needs of the network (“What the world needs”).

Bioregional Zoning

In order for each hub to focus their energies in their immedi-



The current local nodes (Zones 0-2) of the Regenerative network in the center of Italy.

ate vicinity, we use another permaculture framework: Zoning. The twist is that we are applying it to the bioregional network, not just a single site. Liminal Village, our hub, is in “Zone 0.” A few kilometers away we have crowd-funded and created a Waldorf kindergarten together with other local families. That is our bioregional “Zone 1” (*visit daily*), as we and the 22 other families who are in the Kindergarten guild drive to it every day. The Kindergarten guild can be considered as a local distributed community: we have many different Zone 0s (*our different homes*) sharing a Zone 1 for the function of our children’s education. We are “stacking functions” in our daily visits, to satisfy our shared needs, and exchange food, goods, tools, and services at the location. We are now including a food cooperative to provide the network with locally-sourced food staples. We have a veggie box; and also have another guild also on site, which produces flour, chickpeas, pasta, olive oil, and organic wine for our own consumption.

Growing out further to Zone 2 (*visit weekly*), we have acquired a former brick factory and quarry. It is located a 20-minute drive away, near our province’s capital, Ascoli Piceno. The site has building permission for more than 6,000 cubic meters of development. We are planning to use technology to create 3D-printed eco-housing, using locally sourced materials. This site will house the first “ecovillage of services,” which will experiment with a Fair Share economic exchange system between members, with the mission of becoming the bioregional production center for the regenerative network, Regenerativa. At the regional “Zone 3,” and national “Zone 4,” we are contributing by organizing workshops to aid creating local guilds and stack functions of local people and projects, weaving communities and initiatives into a webwork of relations.

Finally, at the global level (Zone 5), we are connected to many incredible individuals, initiatives, villages, and organizations dedicated to change. We are part of the inaugural guild of Regen Civics Alliance, a network of 13 regenerative ecovillages around the world, exploring how to use blockchain technologies for financing and rewarding regenerative contributions across network nodes.

Overall, by framing our contribution in the context of permaculture zones, we are connected at the level of the fractal for each zone. This is a framework that helps us ensure that we are cosmo-locally contributing to the global level, but also able to focus on our immediate surroundings, where we can concentrate on ways to satisfy the needs which are shared with the other nodes in the area.

Conclusion & Call for Action

Permaculture offers a promising path towards regenerative agriculture, human ecologies of mutual support, and sustainable and fair economic systems. By mimicking the principles of nature, and empowering individuals and communities, we can create an economy that is in harmony with the planet and meets the needs of all people. It is time to give permission and support to people to practice permaculture in a diversity of ways, including with the use of digital technologies. It is time for us to

take action and experiment to create increasingly effective and far-reaching permaculture operating systems for a better future for all.

This is a call to action for those willing to experiment with new ways and new narratives, and to create human systems that are an extension of nature, rather than detached from it. Liminal Village is at the forefront of utilizing and embodying the permaculture ethics, principles, and frameworks in the personal, local, and global landscapes. If you are willing to dedicate your knowledge and expertise to the regenerative network so we can all thrive, we invite you to connect and join the regenerative ecology at regenerativa.earth, or to apply to spend a lunation with us at liminalvillage.com.

Roberto Valenti holds a Ph.D. in Artificial Intelligence and Human-Centric Computing. He is using his knowledge and dedicating his time in the research and development of computer-aided community tools that facilitate thriving human ecosystems. His ambition is to combine emerging technologies to create a new operating system that prioritizes the well-being of all living beings, and which supports the creation of regenerative communities. He co-founded the Liminal Village, the first Web3 eco-tech hub and research center in Italy, which researches and experiments with free and open source solutions for meeting human needs. roberto@liminalvillage.com, www.liminalvillage.com, www.regenerativa.earth, www.theos.io

Jillian Hovey holds a B.Sc. in Agriculture and enjoys the grounded complexities of land-based design work and permaculture education. Jillian has been working with eco communities for 30 years, and is a strong advocate of us all cultivating greater awareness and skill in the social landscapes and human ecologies, including that of ourselves. From Montreal, and based outside of Toronto, she has supported people and projects in over 40 countries across 5 continents. Jillian’s permaculture practice is wholistic, and she is honoured to coach and mentor individuals and groups on their regenerative journeys. Jillian@Permaculture.Net, www.JillianHovey.com

Graphics and photos by Roberto Valenti, Laura van Wijn-gaarden, or Jillian Hovey, unless otherwise noted.